ROGERS CORPORATION

\\\ Datasheet

RO4500™ Series
Cost Performance Antenna Grade Laminates

RO4500™ Series High Frequency Laminates are cost/performance materials from Rogers Corporation, specifically engineered and manufactured to meet the specific demands of the antenna markets.

RO4533™ and RO4534™ laminates extend the capabilities of the successful RO4000® product series into antenna applications. This ceramic-filled, glass-reinforced hydrocarbon based material set provides the controlled dielectric constant, low loss performance and excellent passive intermodulation response required for mobile infrastructure microstrip antenna applications.

As with all RO4000® high frequency laminates, RO4500™ laminates are fully compatible with conventional FR-4 and high temperature lead-free solder processing. These laminates do not require special treatment needed on traditional PTFE-based laminates for plated through-hole preparation. This product series is an affordable alternative to more conventional PTFE antenna technologies, thus allowing designers to optimize the price and performance of their antennas. Moreover, RO4533™ and RO4534™ laminates are available halogen-free to meet the most stringent "green" standards.

The resin systems of RO4500™ dielectric materials are designed to provide the necessary properties for ideal antenna performance. The coefficients of thermal expansion (CTEs) in both the X and Y directions are similar to that of copper. The good CTE match reduces stresses in the printed circuit board antenna. The typical glass transition temperature of RO4500™ materials exceeds 280°C (536°F), leading to a low z-axis CTE and excellent plated through-hole reliability. These properties, in combination with a dimensional stability value of less than 0.05%, make RO4500™ laminates an excellent candidate for printed circuit antenna applications. RO4500™ materials also provide increased thermal conductivity over equivalent PTFE/woven glass materials, allowing for design of antennas with increased power handling capability.

In addition to these excellent thermo-mechanical properties, RO4500™ laminates embody electrical characteristics that antenna designers need. The laminates have a dielectric constant (Dk) ranging from 3.3 to 3.5 (±0.08) and a loss tangent (Df) of 0.0020 to 0.0037 measured at 2.5 GHz. These values allow antenna designers to realize substantial gain values while minimizing signal loss. Materials are available with demonstrated low PIM performance, with values better than –155 dBC using two 43 dBm swept tones at 1900 MHz.

\\\ Features and Benefits:

Loss range (0.0020 to 0.0037), Dk range (3.3 to 3.5), low PIM response

- Wide range of application use

Thermoset resin system

- Compatible with standard PCB fabrication

Excellent dimensional stability

- Greater yield on larger panels sizes

Uniform mechanical properties

- Maintains mechanical form during handling

High thermal conductivity

- Improved power handling

\\\ Typical Applications:

- Cellular infrastructure base station antennas
- WiMAX antenna networks

\\\ Standard Properties Table

Properties	Typical Values RO4533™	Typical Values RO4534™	Direction	Units	Condition	Test Method
[1] Dielectric Constant, e_r Process	3.3 ± 0.08	3.4 ± 0.08	Z	-	10 GHz/23°C 2.5 GHz	IPC-TM-650,2.5.5.5
Dissipation Factor	0.0020	0.0022	Z	-	2.5 GHz/23°C	IPC-TM-650, 2.5.5.5
	0.0025	0.0027			10 GHz/23°C	
[2] PIM (Typical)	-157	-157	-	dBc	Reflected 43 dBm swept tones	Summitek 1900b PIM Analyzer
Dielectric Strength	>500	>500	Z	V/mil	0.51 mm	IPC-TM-650, 2.5.6.2
Dimensional Stability	<0.2	<0.3	X,Y	mm/m (mils/inch)	after etch	IPC-TM-650, 2.4.39A
Coefficient of Thermal Expansion	13	11	X	ppm/°C	-55 to 288°C	IPC-TM-650, 2.4.41
	11	14	Y			
	37	46	Z			
Thermal Conductivity	0.6	0.6	-	W/(m.K)	80°C	ASTM C518
Moisture Absorption	0.02	0.06	-	%	D48/50	IPC-TM-650, 2.6.2.1 ASTM D570
Tg	>280	>280	-	°C TMA	A	IPC-TM-650, 2.4.24.3
Density	1.8	1.8	-	gm/cm3	-	ASTM D792
Copper Peel Strength	6.9 (1.2)	6.3 (1.1)	-	lbs/in (N/mm)	1 oz. EDC post solder float	IPC-TM-650, 2.4.8
Flammability	NON FR	NON FR	-	-	-	UL 94
Lead-Free Process Compatible	Yes	Yes	-	-	-	-

[1]Typical values are a representation of an average value for the population of the property. RO4500 LoPro laminates use a modified version of the RO4000 resin system to bond reverse treated foil. Values shown above are for RO4500 laminates without the addition of the LoPro resin. For double-sided boards, the LoPro foil results in a core thickness increase of approximately 0.0007" (0.000018mm) while the resulting Design Dk is dependent on laminate thickness due to LoPro resin Dk being 2.4. Therefore, design Dk is highly dependent on core thickness. For specification values contact Rogers Corporation or visit the Rogers Technology Support hub at http://www.rogerscorp.com.

[2] PIM Performance is heavily influenced by the copper choice. PIM values provided are based on testing of reverse-treat electrodeposited copper foils using Rogers' internal test method on a 0.060" laminate. Typical PIM rating on standard EDC foils are < -145 dBc. Refer to the laminate thickness and copper option table for material options.

Prolonged exposure in an oxidative environment may cause changes to the dielectric properties of hydrocarbon based materials. The rate of change increases at higher temperatures and is highly dependent on the circuit design. Although Rogers' high frequency materials have been used successfully in innumerable applications and reports of oxidation resulting in performance problems are extremely rare, Rogers recommends that the customer evaluate each material and design combination to determine fitness for use over the entire life of the end product.

Standard Thicknesses	Standard Panel Sizes:	Standard Claddings
RO4533 0.020" (0.508mm) +/- 0.0015" 0.030" (0.762mm) +/- 0.0015" 0.060" (1.524mm) +/- 0.0040" **RO4534** 0.020" (0.508mm) +/- 0.0015" 0.032" (0.813mm) +/- 0.0020" 0.060" (1.524mm) +/- 0.0040" *Additional non-standard thicknesses available from 0.020" - 0.060" in varying increments	24" X 18" (610 X 457 mm) 24" X 21" (610 X 533 mm) 24" X 36" (610 X 915 mm) 48" X 36" (1219 X 915 mm) *Additional panel sizes available	*Electrodeposited Copper Foil* *½ oz. (18µm) HH/HH* *1 oz. (35µm) H1/H1* *Additional cladding weights available

*Contact Customer Service or Sales Engineering to inquire about additional available product configurations

RoHS Compliant

\\\ 100 S. Roosevelt Avenue \\\ Chandler, AZ 85226 \\\ Tel: 480-961-1382 \\\ Fax: 480-961-4533 \\\ www.rogerscorp.com

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